

Show all your work in the space provided for each problem. All problems will be graded for correctness, completeness, conciseness, and clarity. Do your best. Don't cut corners.

1. (10 pts) Use induction to prove that $3n+1 < 3^n$ for all $n \geq 2$

① Base case if $n=2$ $3n+1 = 7$ and $3^n = 9$ so $3n+1 < 3^n$ ✓

② Induction Assume for $n \geq 2$ $3n+1 < 3^n$ show $3(n+1)+1 < 3^{n+1}$

$$3(n+1)+1 = (3n+1)+3 \underset{\substack{\uparrow \\ \text{By I.H.}}}{<} 3^n + 3 \underset{\substack{\uparrow \\ 3 < 3^n \text{ for } n \geq 2}}{<} 3^n + 3^n = 2 \cdot 3^n < 3 \cdot 3^n = 3^{n+1}$$

③ Since the inequality holds for $n=2$ and it holds for $n+1$ whenever it holds for n we conclude by math induction that $3n+1 < 3^n \quad \forall n \geq 2$.

2. (6 pts) Prove by exhaustion that $p(x) = x^2 + x$ has four zeros over $\mathbb{Z}_{10}$.

x	0	1	2	3	4	5	6	7	8	9
x^2+x	0	2	6	2	0	0	2	6	2	0

Zeros are 0, 4, 5, 9.

3. a) (4 pts) What does it mean **in terms of limits** to say the function f is continuous at a number a in its domain?

f is continuous at a means $\lim_{x \rightarrow a} f(x) = f(a)$

b) (4 pts) State carefully what this means **in terms of ϵ and δ** .

$\forall \epsilon > 0 \exists \delta > 0$ such that if $|x - a| < \delta$ then
 $|f(x) - f(a)| < \epsilon$.

4. (10 pts) Use Cauchy's definition of limit to prove that $\lim_{x \rightarrow 2} x^2 + x - 10 = -4$.

First assume $\delta = 1$ so $|x - 2| < 1 \Rightarrow 1 < x < 3$

$$|f(x) - L| = |x^2 + x - 6| = |(x+3)(x-2)| = |x+3| \cdot |x-2| < 6|x-2| \text{ for } 1 < x < 3$$

Let ϵ be any positive real number and choose $\delta = \min \left\{ 1, \frac{\epsilon}{6} \right\}$

so $0 < |x - 2| < \delta$ means $|x - 2| < \frac{\epsilon}{6}$ and $|x - 2| < 1$ and $|x + 3| < 6$

$$\therefore |f(x) - L| = |x + 3| \cdot |x - 2| < 6 \cdot \frac{\epsilon}{6} = \epsilon$$

$$\therefore \lim_{x \rightarrow 2} x^2 + x - 10 = -4$$

5. a) (5 pts) What does it mean to say that a is algebraic over $\mathbb{Q}$?

a is a zero for some polynomial $p(x) \in \mathbb{Q}[x]$.

b) (5 pts) Prove that $3 + \sqrt{5}$ is algebraic over $\mathbb{Q}$.

$$w = 3 + \sqrt{5}$$

w is a zero for $p(x) = x^2 - 6x + 4 \in \mathbb{Q}[x]$.

$$w - 3 = \sqrt{5}$$

$$w^2 - 6w + 9 = 5$$

$$w^2 - 6w + 4 = 0$$

6. (8 pts) Let A and B be countably infinite sets that have no elements in common. Prove that the union of A and B will be countably infinite.

$$A = \{a_1, a_2, a_3, \dots\}$$

$$A \cup B = \{a_1, b_1, a_2, b_2, a_3, b_3, \dots\}$$

$$B = \{b_1, b_2, b_3, \dots\}$$

Define a bijection from $\mathbb{N}$ to $A \cup B$ by

$$\begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \dots \\ f \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ a_1 & b_1 & a_2 & b_2 & a_3 & b_3 & a_4 \dots \end{array}$$

$$f(i) = a_{\frac{i+1}{2}} \quad \text{if } i \text{ is odd}$$

$$f(i) = b_{\frac{i}{2}} \quad \text{if } i \text{ is even.}$$

7. (12 pts) Prove that every linear function $f(x) = mx + b$ with $m \neq 0$ is one-to-one and find the inverse of the function.

One-to-one $f(x_1) = f(x_2) \Rightarrow mx_1 + b = mx_2 + b$

$$\therefore (mx_1 + b) - (mx_2 + b) = 0$$

$$\therefore m(x_1 - x_2) = 0$$

But $m \neq 0$ so $x_1 - x_2 = 0$

$$\therefore x_1 = x_2$$

$$f(f^{-1}(x)) = x$$

$$m \cdot f^{-1}(x) + b = x$$

$$m \cdot f^{-1}(x) = x - b$$

$$f^{-1}(x) = \frac{x-b}{m} \quad \text{or} \quad \frac{1}{m}x - \frac{b}{m}$$

8. (9 pts) Let G be the group $U(7)$ under multiplication mod 7. Let f be the function from G to G that maps every element of G to its multiplicative inverse.

a) (6 pts) Fill in all missing values in the table.

x	1	2	3	4	5	6
$f(x)$	1	4	5	2	3	6

b) (3 pts) Circle all answers that accurately complete the sentence:

The function f in part a is

an injection

a surjection

a bijection

a rejection

9. (10 pts) Let f be any function that is positive valued and differentiable at $x = c$.

a) According to the definition, what is the derivative of f at c ?

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

b) Use the definition of the derivative to find the formula for the derivative of $(f(x))^2$ at $x = c$.

$$f'(c) = \lim_{x \rightarrow c} \frac{[f(x)]^2 - [f(c)]^2}{x - c} = \lim_{x \rightarrow c} \frac{[f(x) - f(c)][f(x) + f(c)]}{x - c}$$

$$= \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \cdot \lim_{x \rightarrow c} [f(x) + f(c)] = f'(c) \left(\lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} f(c) \right)$$

$$= f'(c) \cdot (f(c) + f(c))$$

$$= 2 f(c) \cdot f'(c)$$

10. (9 pts) Use one of Cantor's diagonalization arguments to prove that the set of all real numbers in the interval $(0,1)$ is uncountable.

Assume for contradiction that the interval $I = (0,1)$ is countable

If so we can make a list that contains all numbers in I .

$$a_1 = 0.a_{11}a_{12}a_{13}\dots$$

$$a_2 = 0.a_{21}a_{22}a_{23}\dots$$

$$a_3 = 0.a_{31}a_{32}a_{33}\dots$$

$\vdots$

Define number $d = 0.d_1d_2d_3\dots$ where

$$d_i = 0 \text{ if } a_{ii} \neq 0 \text{ and } d_i = 1 \text{ if } a_{ii} = 0.$$

The number $d \in I$ but $d \neq a_i$ for any i because d and a_i differ in the i^{th} decimal place.

$\therefore$ There can be no complete list

$\therefore I$ is not countable

$\therefore I$ is uncountable

11. (8 pts) Give examples of the following.

many examples for all of these.

a) a real number that is not algebraic over $\mathbb{Q}$

$$\pi \text{ or } e$$

b) a real number that is algebraic over $\mathbb{Q}$ but is not rational

$$\sqrt{2}$$

c) a function from $\{1,2,3,4\}$ to $\{5,6,7\}$ that is a surjection



$$\{(1,5), (2,5), (3,6), (4,7)\}$$

d) a function from $\{1,2,3,4\}$ to $\{5,6,7\}$ that is **not** a surjection



$$\{(1,5), (2,5), (3,5), (4,6)\}$$